UNIVERSITÀ di VERONA

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HCI – intelligent *multimodal* interfaces

Interaction&Usability

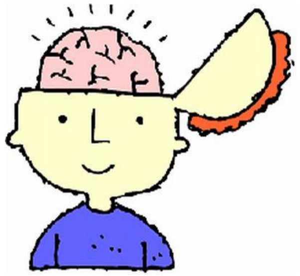
Master program in Artificial Intelligence

Interaction design

- Human factors
- Appearance and presentation,
- Interaction

Human factors

- Identifying and understanding of human **characteristics** and **limitations** involved in the user activities to solve an interaction task:
  - perception,
  - memory (e.g., short term memory),
  - attention and learning



Human factors

- **Two principles**:
    1. Give **control** to the user,
    2. **Reduce** the amout of information to be **reminded** by the user

User control

- The user should guide and control the interface,
- The user decides which function to activate,
- The user decides the information to be shown, how and when…, also the aesthetic aspects…,

- **Flexible interface**: the interface should differentiates wrt different users, experties, abilities and so on…

  E.g.,
  - give the user different options to solve the same task,
  - Interface customization

Reduce the amount of information to be reminded by the user

- It is easier for a person to ***recognize*** than ***remember***
  - E.g., recognize a command among a list of options rather than remember the name

- **Good practice**:
  - Show the user all the available commands and options,
  - Show the proper information to complete a certain process,
  - Organize the information to help the user to focus on the essential information to solve a task

Appearance and presentation

- Visual characteristic of the interaface: layout, colors, shape and dimension of the involved elements...
- **Three principles:**
    1. Create an attractive aspect,
    2. Use significative and recognizable object representations,
    3. Keep the consistances of the interface

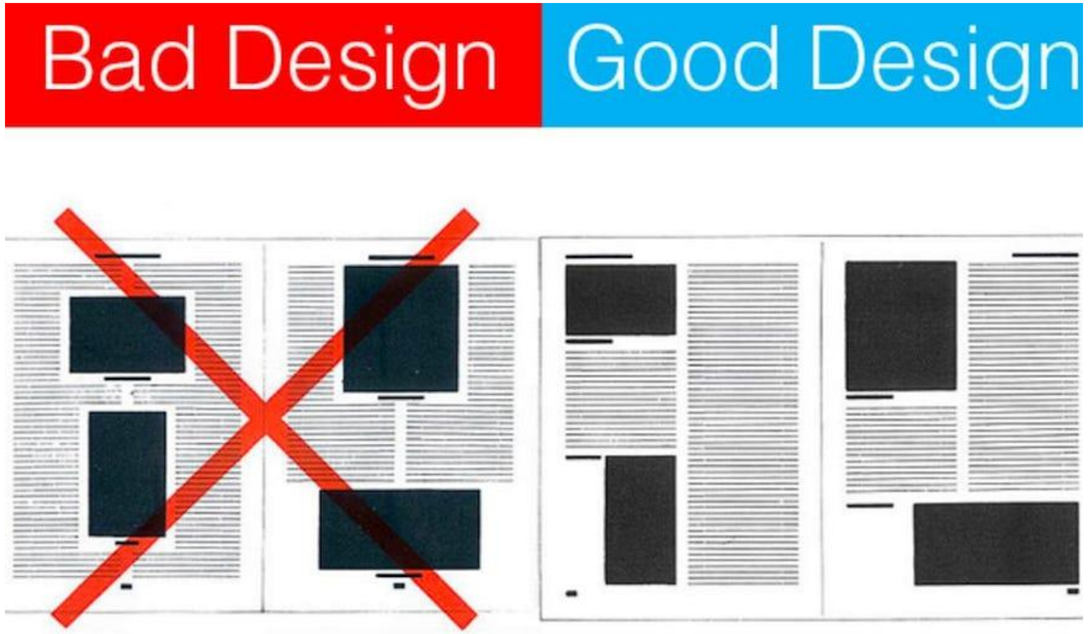
Attractive aspect

- All elements on the screen should be visible and undertandable,
- A proper use of contrast helps,
- Colors, shapes, space, organization of the objects in the screen

Bad Examples	Good Examples
Blue on black is bad	Yellow on black is good
Green on orange is bad	Black on orange is good
Red on green is bad	Black on green is good
Grey on purple is bad	White on purple is good

https://www.csun.edu/universal-design-center/web-accessibility-criteria-color-contrast

Attractive aspect



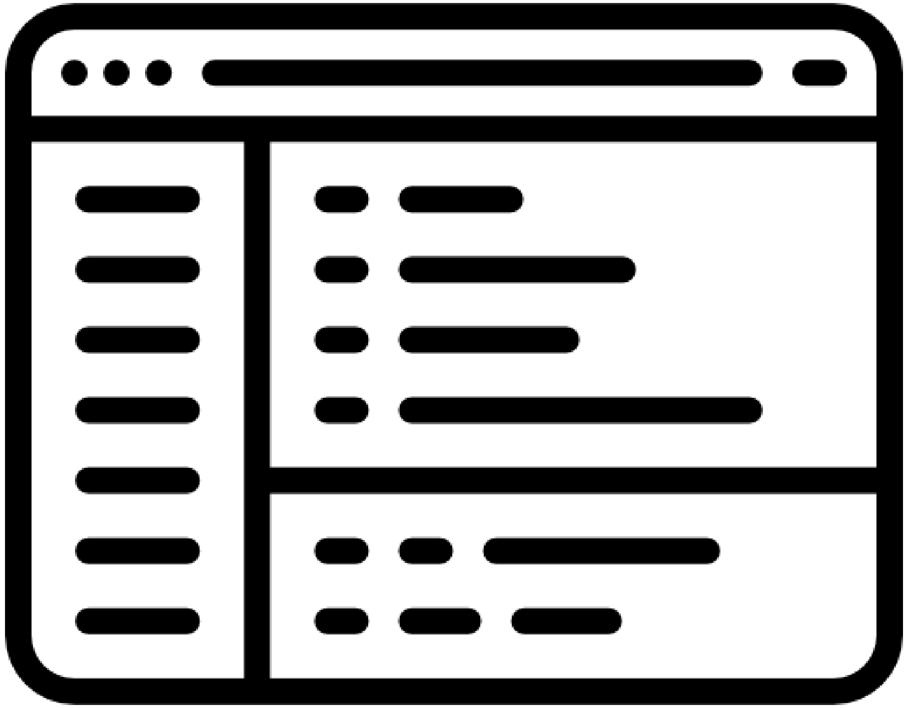
bad contrast

good contrast

https://www.mockplus.com/blog/post/bad-ui-design-examples

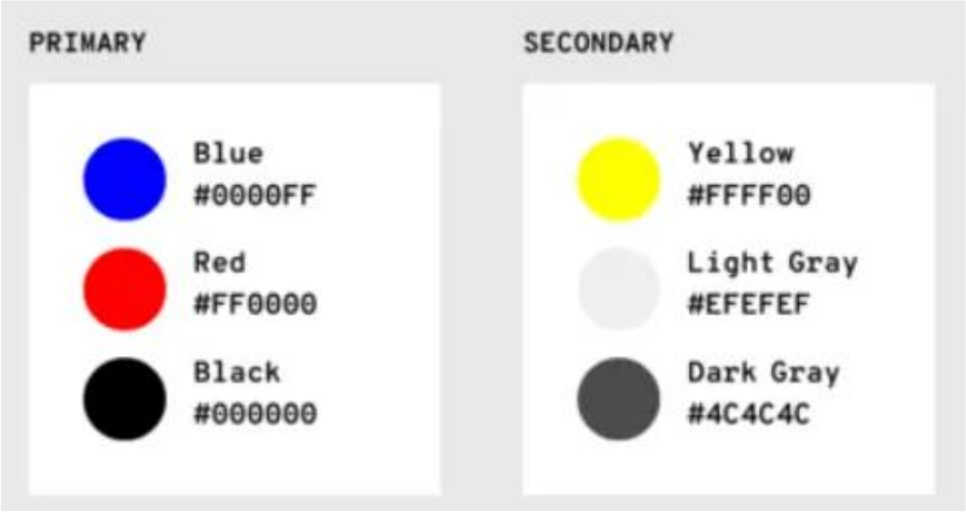
Object representation

- Object representation must be **clear**, **significative** and **distinguishable** from other objects,
- A good practice is to **group** (hierarchically) the interface elements in class (i.e., affinity class or functionality class),
  - A **visual coherece** should be employed to associate each object to the correct class



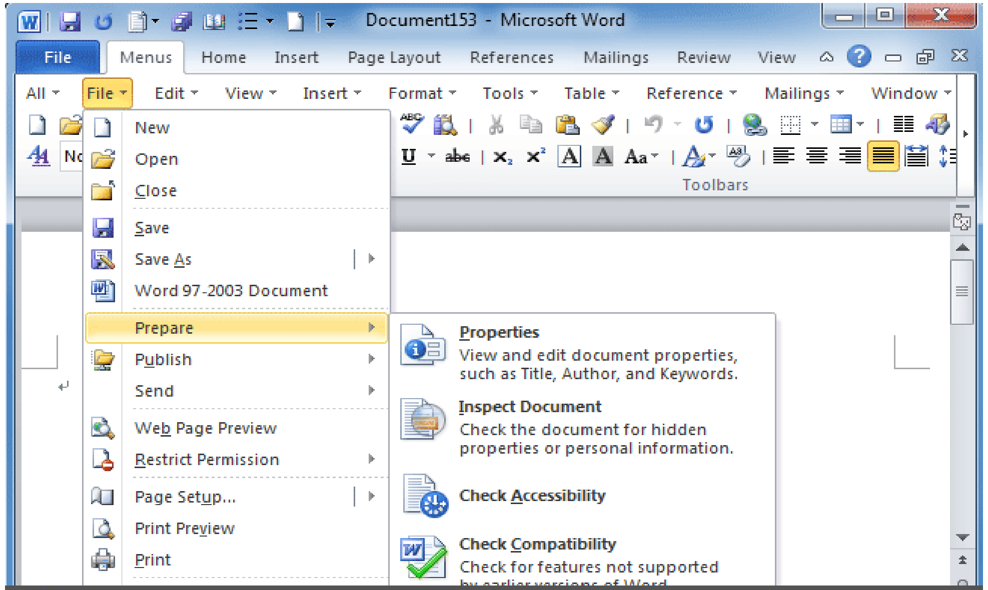
Consistency

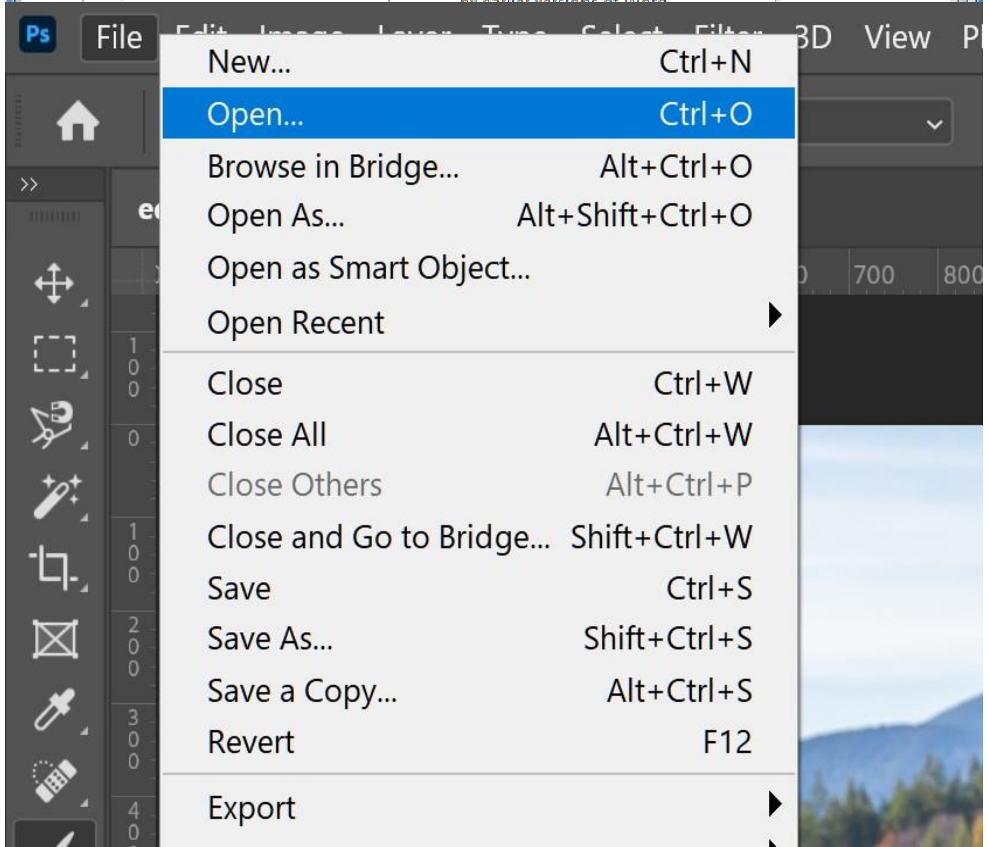
- **Visual consistency:**



Consistency

- **Functional consistancy**: the same action shold give the same results in different applications
  - E.g., Open, close, save





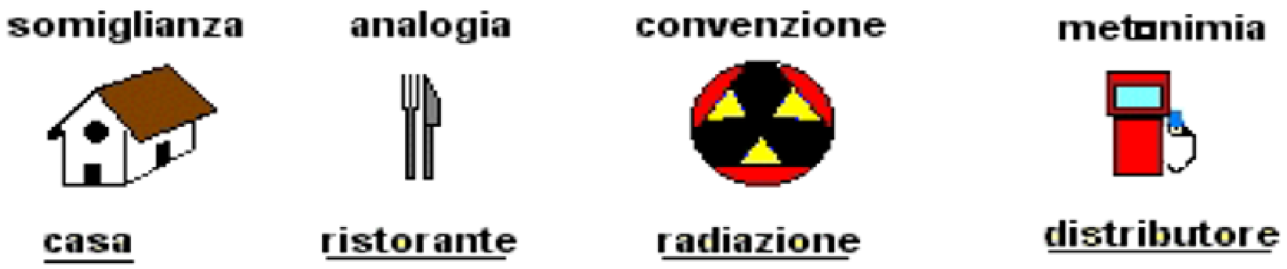
Consistency

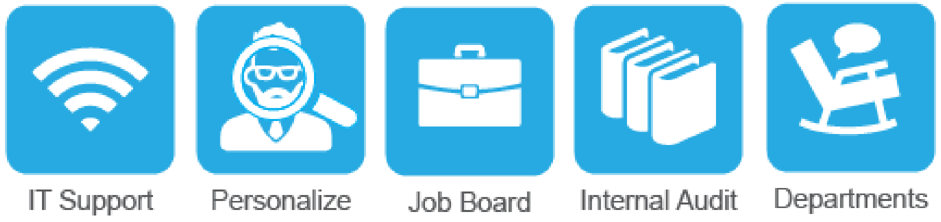
- A **consistence** design provides a more **predictable** interface:
  - visual information and visual aspect,
  - how to manipulate the information,
  - navigation methods

- Reuse of interface components
- Easier learning for the user

Icons

- Icons should suggest how to use the interface,
- Non-consistent icons can cause misunderstanding





Examples of intranet icons that are not helpful (from left to right): the ubiquitous wireless symbol used here to represent support, a hard-to decipher image of a person with a magnifying glass over his face to represent personalization, a briefcase standing for job listings, books to represent internal audit, and a chair with a speech bubble to represent departments.

https://www.nngroup.com/articles/bad-icons/

Interaction

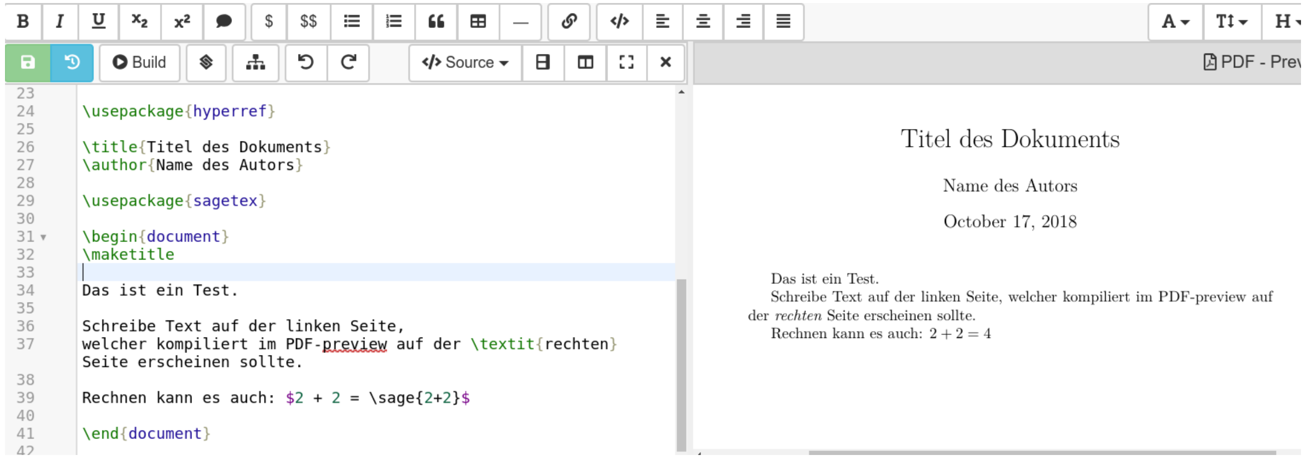
- The way the user controls the execution of an application,
- Interaction can involved:
    - keyboard,
    - mouse,
    - touch devices

Three design principles:

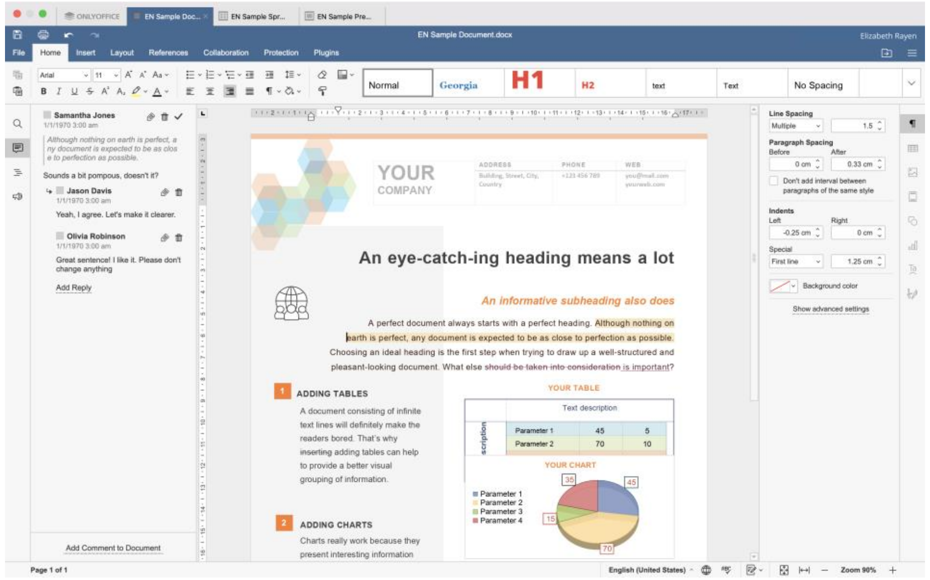
- Direct handling,
- Feedback,
- Error tolerancce

Direct handling

- The user directly acts to an object (e.g., an icon)
  - The aim is to mimic the natural way to interact with real objects (e.g., press a button),
- The interactive paradigme «What you see is what you get (WYSIWYG)» (e.g., text editor)



Latek



word

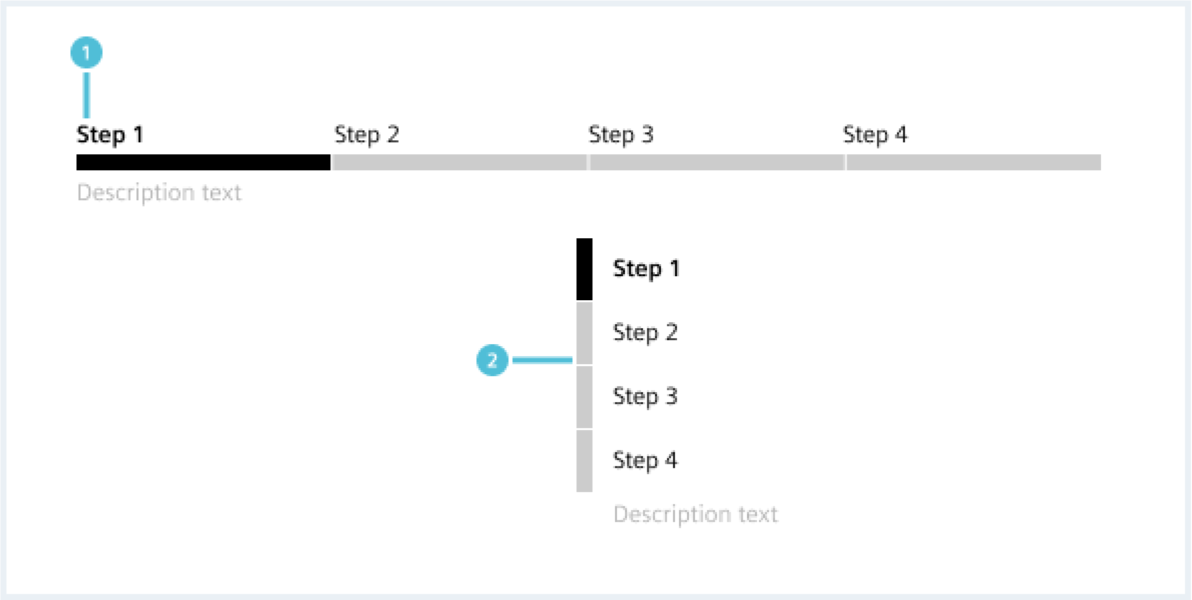
Feedback

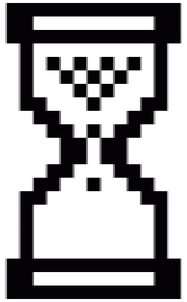
- The interface should properly react to the user actions
  - The user should receive an immediate feedback from the interface, this improves the user awareness and learning

E.g.,:

- highlights an element once it is selected by the user,
- make a sound when a button is pressed,
- change the shape of the pointer to show the state of the application (e.g., an hourglass for waiting).

Feedback



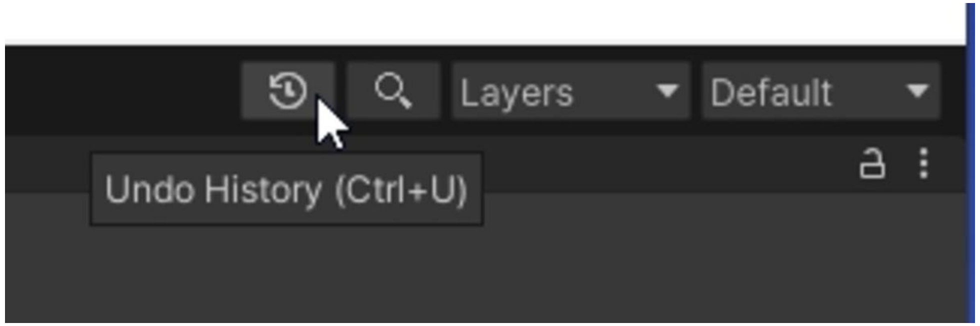


https://design.mindsphere.io/patterns/progressbar.html

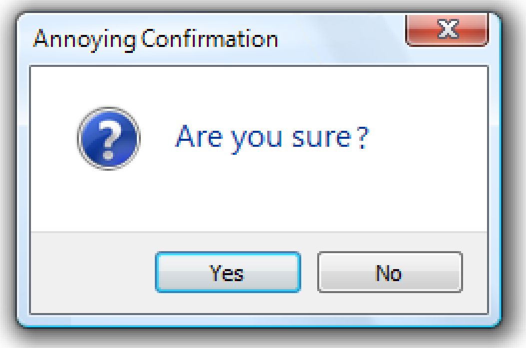
Error tolerance

- The system can accept wrong actions of the user without negative consequences.

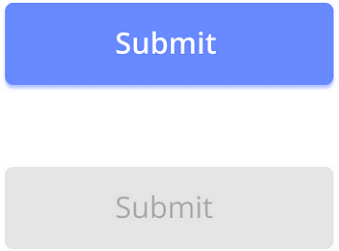
Common practices



Undo and undo history



Confirmation window



Disabled buttons

Other design principles

- Affordance,
- Causality,
- Constraints,
- Conceptual models,
- Visibility,
- Mapping

Affordance

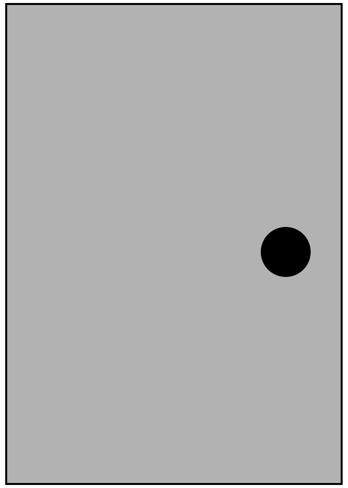
- Affordance is used to help the user in using the system,
- Real properties of an object that are perceived from the user suggest the user how to use the object
  - For the user it is sufficient to observe the object:

A knob suggests to turt it

A button suggests to press it

A keyhole suggests to insert the key

Affordance



Push or pull?

OK push, but in which side?

correct

Affordance


PUSH
NO!


OK!


1
ATM
AZIENDA TRASPORTI MILANESI S.p.A.
AZIENDA TRASPORTI MILANESI
LINEE ORDINARIE URBANE
0.989.744
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2
OK
CACCIA IN MEZZO A MILANO
PARTECIPA ANCHE TU ALLA GRANDE CACCIA AL TESORO ATM
UN EVENTO DA NON PERDERE!
LINEE ORDINARIE URBANE
0262838
004
Davide 'Folletto' Casali
im.digitalhymn.com

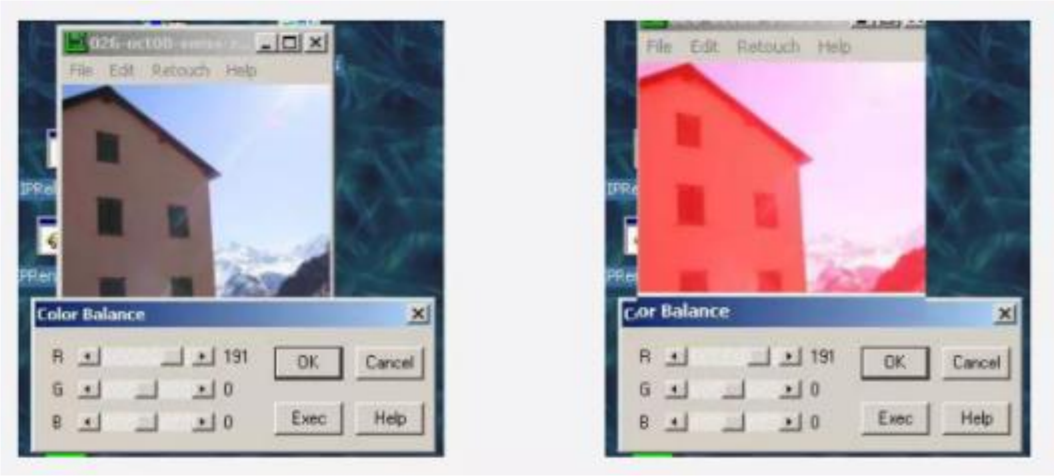

NO!
3
AZIENDA TRASPORTI MILANESI S.p.A.
ATM
LINEE ORDINARIE URBANE
017
0722360
Davide 'Folletto' Casali
im.digitalhymn.com

Causality

- When an action appears right after another action the user assigns it a causality relation,
  - Interpretation of «feedback»,
  - False causality,
  - Incorrect effect,

Overshooting

- Effects visible only after Exec button is pressed
- OK does nothing!
- Awkward to find appropriate color level

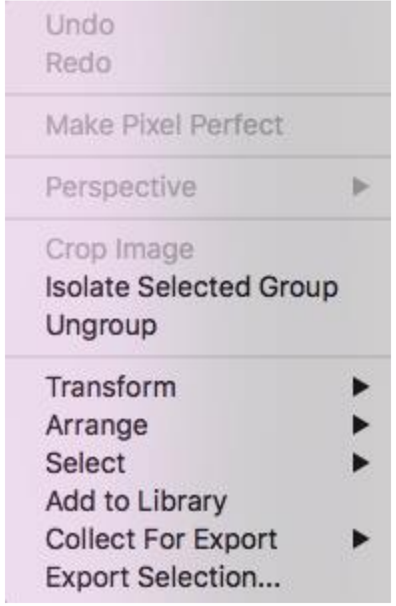


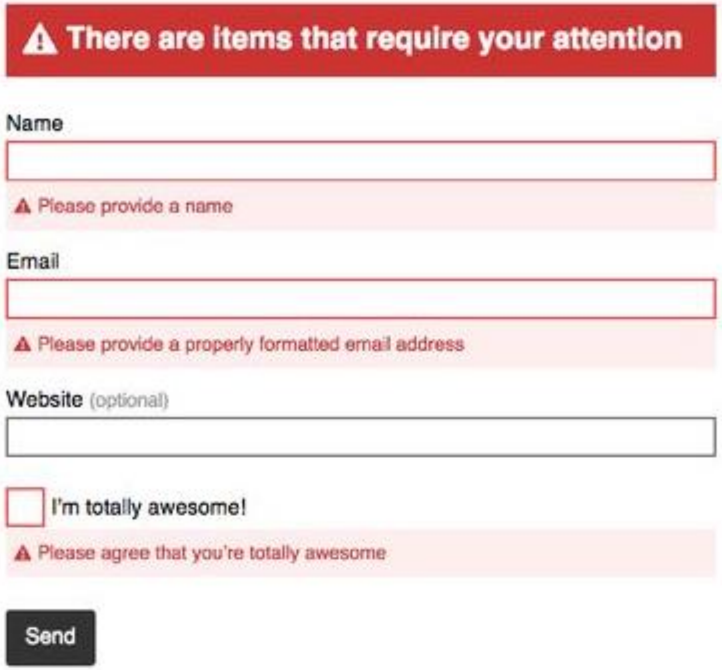
No apparent cause-effect relation

Constraints

- Constraints reduce the freedom of the user,
  - **Physical** constraints: i.e., scissor,
  - **Cultural** constraints and **conventions**,

    Red means danger, **green** means safe





cash machine

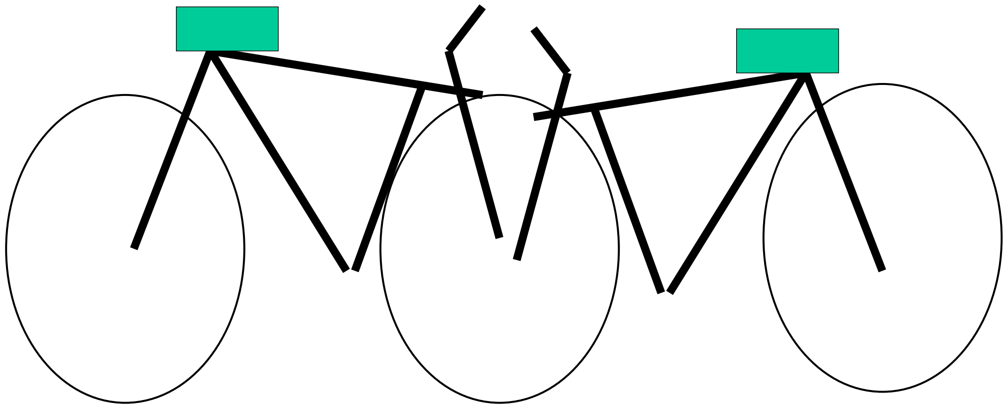
calculator

Conceptual model

- People have a «mental models» of how things work,
- Conceptual models come from user experience, learning and instruction

Conceptual model

- The mental model of a device is composed by the interpretation of its perceived acgtions and its visible structure
- When an object or a device is observed the user defines a mental model on how the object behaves,
- The user uses this mental model to simulate the behaviour of this object



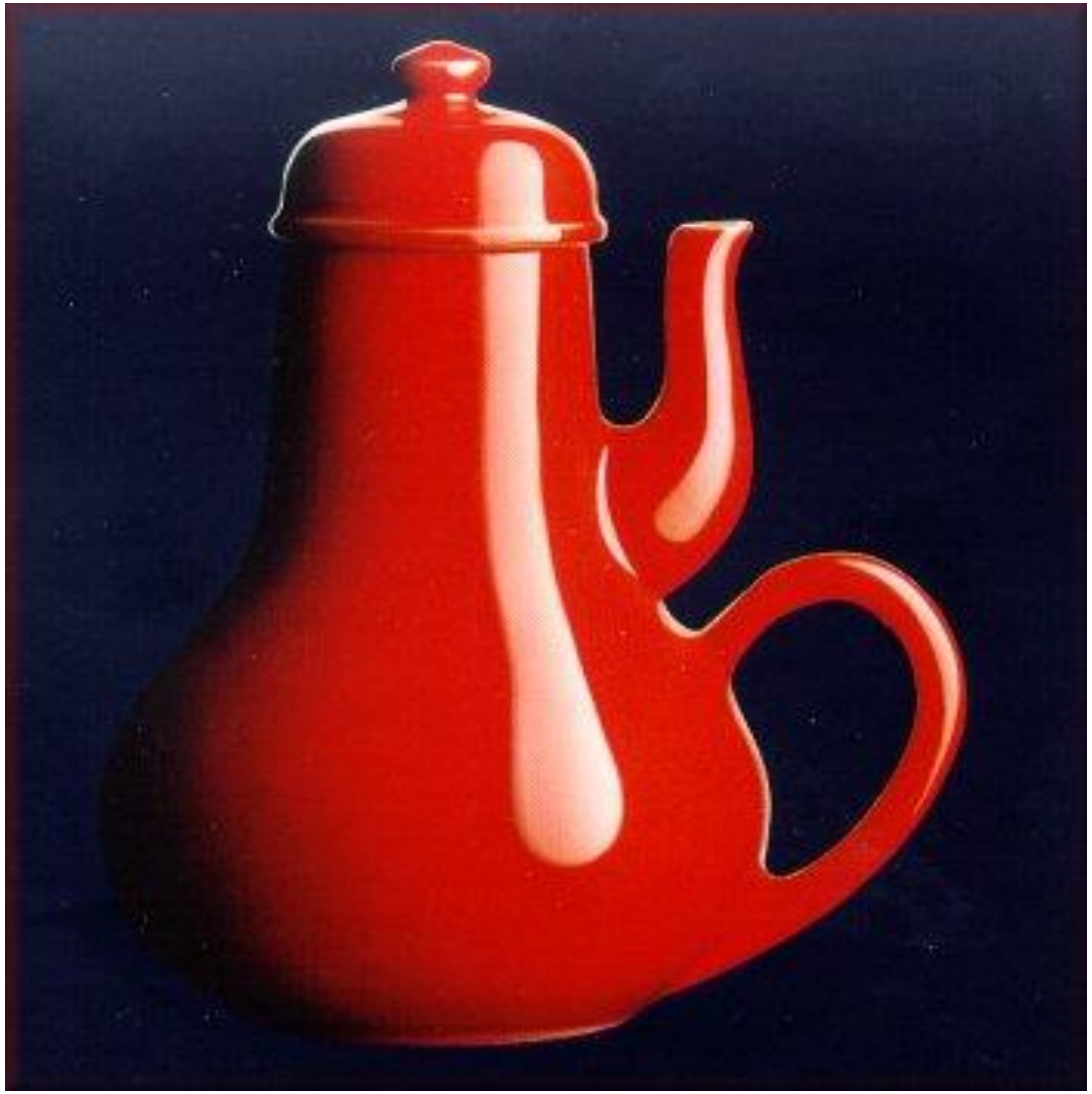
Conceptual model

- A **good** conceptual model enables the users to predict the effects of their actions,
- A **good** conceptual model enable the user to understand the relations between the device controls and outputs.

- A **bad** conceptual model enforces the user to operate blindly,
- A **bad** conceptual model makes the identification of action effects more difficult,
- A **bad** conceptual model makes the addressing of new situations more difficult.

Conceptual model

Is this coherent with the conceptual model?



Visibility

- Visibility is achieved by placing the controls in a highly visible location:
  - The more visible functions are, the more likely user will be able to know what to do next
- Design for visibility means that just by looking, users can see the possibilities for action,
  - To indicate what parts operate and how,
  - To indicate how the user is to interact with the device

Visibility

- Visibility is often violated in order to make thinks «look good»,
- Problems arise when the number of possible actions are higher than visible controls (i.e., some functions becomes invisible)

Control panel of an elevator

... you need to insert your room card in the slot by the buttons to get the elevator works!

http://www.baddesigns.com/

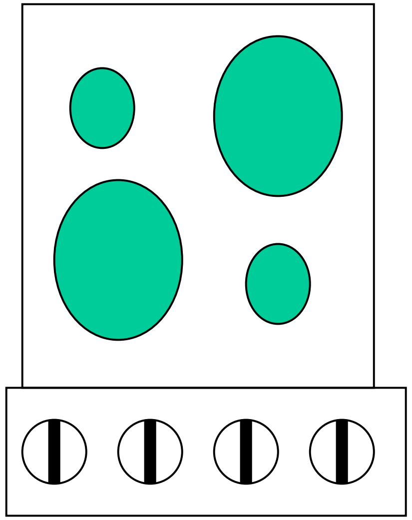
Mapping

- A mapping is a relation between two obects,
- If the mapping is easy to learn and remember also the use of device becomes easier…,
- Natural mappping are effective:
  - Spatial mapping,
  - Perceptual mapping (i.e., bigger means stronger)

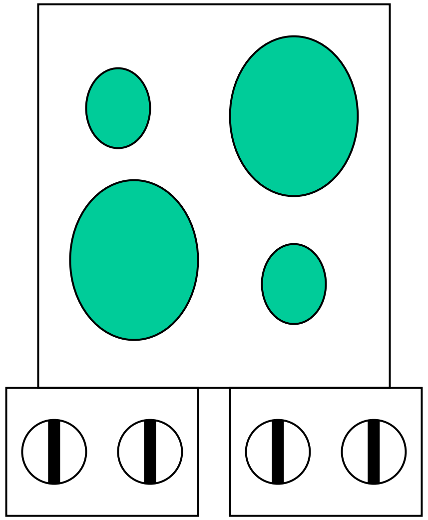
A device is easy to use when possible actions are visible and controls exploits natural mapping

Mapping

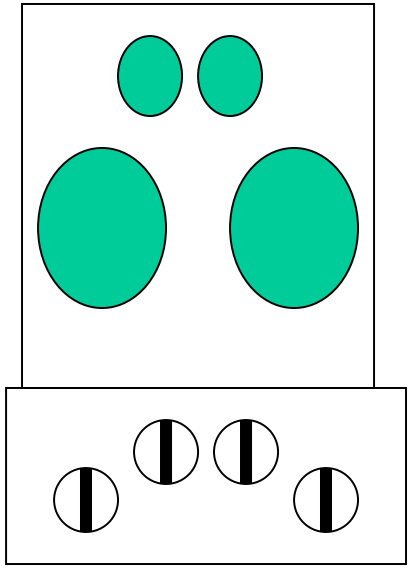
Arbitrary, 24 options



Pairwise: 4 options



Complete mapping



Mapping

https://en.wikipedia.org/wiki/Natural_mapping_(interface_design)

The placement of controls for adjusting the positioning of a car seat is extremely unintuitive. The intention behind the vertical and horizontal shaped controls are to reflect the movement of the seat; however, there is no indication to move the controls in the intended ways without referring to the image labels. This is a poor design because the controls are placed on the side of the seat, which is not visible to users when they are driving. Thus, the user must go through many trial and error attempts to figure out which control moves the seat forward, backward, upright, or laying flatter. There are also many other additional buttons that are arbitrarily placed next to one another with no tactile [feedback] on the controls themselves to indicate their functionalities.

In this example, the placement of controls for adjusting the positioning of a car seat is more intuitive and easier to use because the arrangement of controls directly mirrors the shape of a real car seat. This is especially useful during the process of driving when it is impossible to read the labels on the controls because the user can easily operate the controls without having much prior knowledge of each control's exact functionality. The bottom button clearly moves the bottom part of the seat forward of backward. The top button maps to the backrest of the car seat and dictates the vertical orientation of that part of the seat, moving it either more upright or flatter. This presentation of controls greatly aids the user in better understanding the state of the system and figuring out how to achieve their goal of adjusting their car seat to their liking without much cognitive strain.

Historical evolution

Classic interaction

- Traditional computer interfaces are based on the «**Desktop**» concept as an example of «**remediation**»

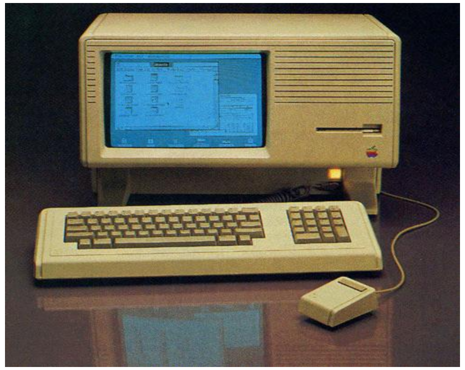
Jay David Bolter and **Richard Grusin:**
"new visual media achieve their cultural significance precisely by paying homage to, rivaling, and refashioning such earlier media as perspective painting, photography, film, and television. They call this process of refashioning "**remediation**," and they note that earlier media have also refashioned one another: photography remediated painting, film remediated stage production and photography, and television remediated film, vaudeville, and radio".

Classic interaction

Traditional model of HCI interface: desktop methaphore, icons, screen, keyboard, mouse

1981 Xerox Star workstation:
first GUI generation (Graphical User Interface),

1984 Apple Macintosh GUI: new style of HCI (Human-Computer Interaction)



Pervasive interfaces

From **desktop** interface to **pervasive** interface:

- The computational power is moved in the environment
- The interface identifies and serves the user

“Ubiquitous computing names the third wave in computing, just now beginning. First were mainframes, each shared by lots of people. Now we are in the personal computing era, person and machine staring uneasily at each other across the desktop. Next comes ubiquitous computing, or the age of calm technology, when technology recedes into the background of our lives."

Mark Weisen

"The Computer for the 21st Century" - *Scientific American Special Issue on Communications, Computers, and Networks*, September, 1991
http://www.ubiq.com/hypertext/weiser/SciAmDraft3.html

Pervasive computing

- **Similar names**:
  - Ubiquitous Computing (Mark Weiser, 1991)
  - The Invisible Computer (Donald Norman, 1998)
  - Disappearing Computer (European IST, 2000)

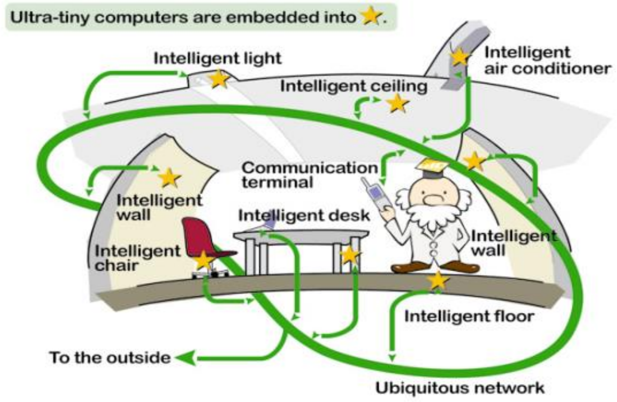
Or:

Appliance Computing, Situated Computing, Ambient Intelligence, Calm Computing, Ambient Displays, Context-Aware Computing

Pervasive computing

- Computation is part of our **daylife**, standard objects become **computational objects** as well,

- Idea of “***calm computing***”: the computer does not absorb the user's full attention (like a movie), but requires an ***intermittent*** level of attention (like the radio)



Pervasive computing

- Interfaces must be designed in a distributed and reactive way with respect to user behaviors, it is the user who guides the interaction.

- Balance between **center** and **periphery**: computation is integrated with other activities that the user must carry out

"calm technology aims to reduce the excitement of information overload by letting the user select what information is at the center of their attention and what information is peripheral", Mark Waiser

Pervasive computing

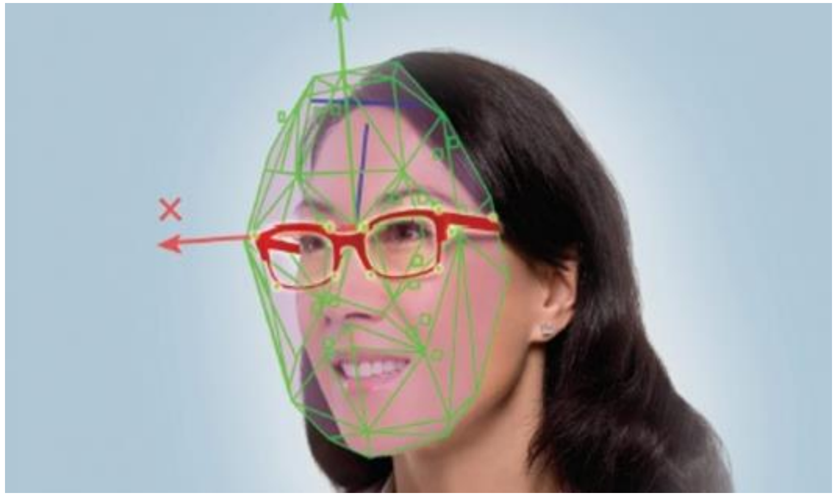
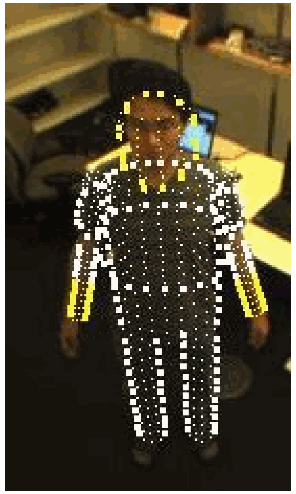
The user interaction with the machine changes:

- **Implicit input**: these are human actions and behaviors that are carried out to achieve a goal. They are not intended primarily as interactions with computers, but they are captured, recognized and interpreted by computers as input

- **Implicit output**: computer output that is not related to explicit input and that is integrated with the environment and user goals in a transparent manner

Pervasive computing

- From **interactive interfaces** to **perceptive interfaces**
- **Recognition** procedures are crucial


The computer must understand the user's intentions by automatically acquiring various information about the user through a combination of sensors

Pervasive computing

Pervasive computing refers to other more specific concepts to highlight particular aspects:

- **Wearable Computing**: computation in cloths, the user wears computable objects
- **Tangible Computing** (Ishii): computation in everyday objects, the world as an interface (TUI vs GUI),
- **Locative Media** (Karlis Kalnins): additional information made available and adapted in specific places with the aim of enhancing the place itself
- **Intelligent Environments**: computing integrated into the environment and distributed. The environment responds to the movements and needs of users

Tecnological tools

- **Just to mention a few…..**

  wifi, www, socialnetwork, QR code, RFID, GPS, GoogleMap, StreatView, dispositivi testuali e sistemi di riconoscimento del testo, strumenti touch screen, tabletPC, data glove, eyegaze, dispositivi olografici, HMD, optical see thorough, video-based, CAVE, work banch, TOF cameras, stereo cameras, ….

Involved infrastructures

Hardware infrastructures

Internet

Satellite network

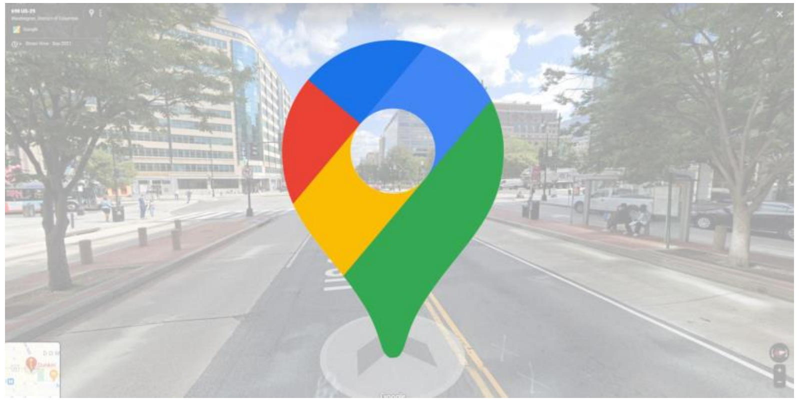
Computational power

Interaction devices

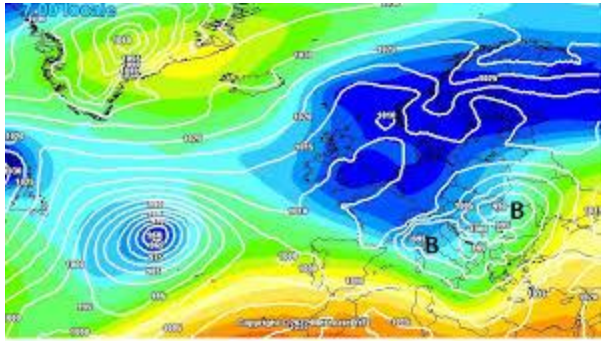
Involved infrastructures

Data&Software infrastructures

PyTorch



OpenStreetMaps



Involved infrastructures

Methodological infrastructures

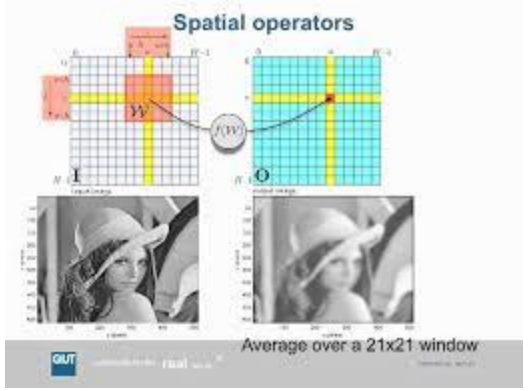
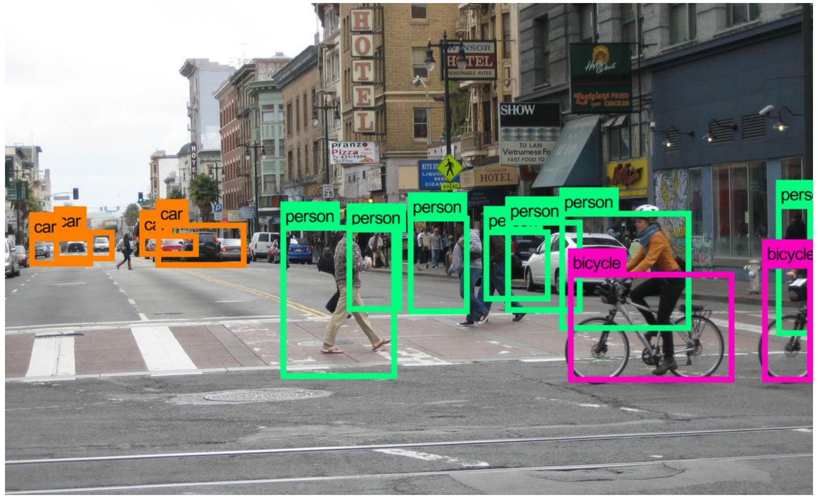
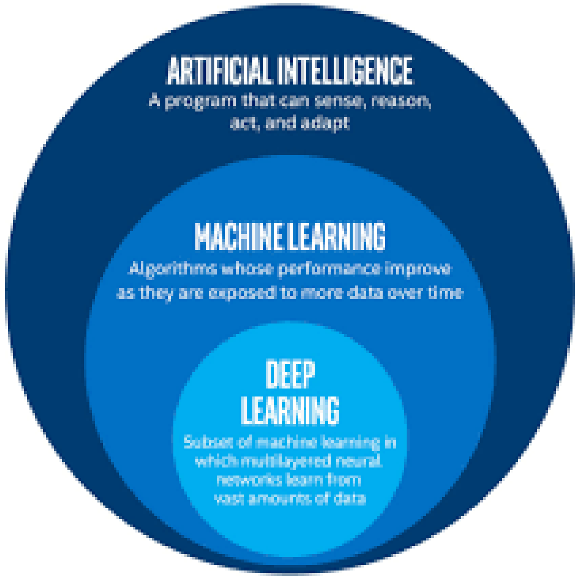


Image processing



Computer Vision



Take home message

- Recent scientific and tecnological advances allow the machines to solve a large amount of tasks,
- The user must be confortable in the interaction with the machine,

The HCI design must consider the human characteristics and behaviours

Take home message

- **Natural interfaces** are more effective for the user since the user is focused on the solution of the task rather than understaning how to interact with the machine (i.e., the machine disappears)
- The idea of **pervasive interaction** is nowadays feasible due to the large amout of tecnological solution and infrastructures,

The HCI solutions must exploit AI and machine learning techniques to understand the human